

DPN2 – DPN2 TASK 2: CLOUD DATA SOLUTION IMPLEMENTATION

CLOUD DATABASES – D607

PRFA – DPN2

[Preparation](#)[Task Overview](#)[Submissions](#)[Evaluation Report](#)

COMPETENCIES

4167.1.3 : Implements a Cloud Data Storage Structure

The learner implements a cloud data storage structure to address organizational requirements.

INTRODUCTION

In this task, you will demonstrate your ability to implement the planned cloud data storage design you created in Task 1.

You will use the information provided in the "Alliah Company Scenario," "Data Dictionary for Transaction Data," and "Transaction Data," documents to respond to the prompts for this task.

You will complete this performance assessment in the provided WGU GCP Database Playground.

SCENARIO

After convincing Alliah Company to move their data from an on-premises solution to a cloud data solution in Task 1, you are now responsible for creating a sandbox environment to demonstrate that the cloud storage solution works as intended.

Use the provided "Alliah Company Scenario," "Data Dictionary for Transaction Data," and "Transaction Data" documents to complete these tasks:

- Create the required database schema in the Google Cloud Platform (GCP) data store that you identified as optimal in Task 1.
- Populate the schema with records from the sample transaction data.
- Perform a query demonstrating that the data store functions as designed.

REQUIREMENTS

Your submission must represent your original work and understanding of the course material. Most performance assessment submissions are automatically scanned through the WGU similarity checker.

Students are strongly encouraged to wait for the similarity report to generate after uploading their work and then review it to ensure Academic Authenticity guidelines are met before submitting the file for evaluation. See [Understanding Similarity Reports](#) for more information.

Grammarly Note:

Professional Communication will be automatically assessed through Grammarly for Education in most performance assessments before a student submits work for evaluation. Students are strongly encouraged to review the Grammarly for Education feedback prior to submitting work for evaluation, as the overall submission will not pass without this aspect passing. See [Use Grammarly for Education Effectively](#) for more information.

Microsoft Files Note:

Write your paper in Microsoft Word (.doc or .docx) unless another Microsoft product, or pdf, is specified in the task directions. Tasks may not be submitted as cloud links, such as links to Google Docs, Google Slides, OneDrive, etc. All supporting documentation, such as screenshots and proof of experience, should be collected in a pdf file and submitted separately from the main file. For more information, please see [Computer System and Technology Requirements](#).

You must use the rubric to direct the creation of your submission because it provides detailed criteria that will be used to evaluate your work. Each requirement below may be evaluated by more than one rubric aspect. The rubric aspect titles may contain hyperlinks to relevant portions of the course.

- A. Using the instructions provided in the WGU GCP Database Playground, do the following:
1. Create the required schema objects within the appropriate Google Cloud Platform database, including tables, collections, documents, or index structures as appropriate that incorporate all the attributes from the dataset. Submit a screenshot of your schema as it appears in the GCP database.
 - a. Submit documentation of any code you used to create the database schema in part A1, identifying the components of the code and how they function.
- B. Using the instructions provided in the WGU GCP Database Playground, populate your database within the Database Playground by doing the following:
1. Insert at least **10** complete transaction records from the sample transaction dataset into your database. Submit a screenshot from the GCP database showing the inserted records.
- Note: This can be accomplished manually or by using scripting and programming approaches.*
2. Explain the components of the code used in part B1 and how they function. Submit documentation of the code used to insert your records.
- C. Provide a Panopto video recording that includes *both* a screen share of the presenter demonstrating the following queries running within the GCP database, and a verbal explanation of how the queries below were written.
1. Write and execute a query listing *all* of the unique customers in your database.
 2. Write and execute a query listing *all* of the items in a single customer's shopping cart.
 3. Write and execute a query listing the total purchase amounts for *all* customers in the database in descending order.

Note: For instructions on how to access and use Panopto, use the "Panopto How-To Videos" web link provided below. To access Panopto's website, navigate to the web link titled "Panopto Access" and then choose to log in using the "WGU" option. If prompted, log in using your WGU student portal credentials, and then it will forward you to Panopto's website.

To submit your recording, upload it to the Panopto drop box titled "Task 2: Cloud Data Solution Implementation – DPN2 | D607". Once the recording has been uploaded and processed in Panopto's system, retrieve the URL of the recording from Panopto and copy and paste it into the "Links" option. Upload the remaining task requirements using the "Attachments" option.

D. Acknowledge sources, using in-text citations and references, for content that is quoted, paraphrased, or summarized.

E. Demonstrate professional communication in the content and presentation of your submission.

File Restrictions

File name may contain only letters, numbers, spaces, and these symbols: ! - _ . * ' ()

File size limit: 200 MB

File types allowed: doc, docx, rtf, xls, xlsx, ppt, pptx, odt, pdf, csv, txt, qt, mov, mpg, avi, mp3, wav, mp4, wma, flv, asf, mpeg, wmv, m4v, svg, tif, tiff, jpeg, jpg, gif, png, zip, rar, tar, 7z

RUBRIC

A1:GCP DATABASE SCHEMA

NOT EVIDENT

A screenshot showing a Google Cloud Platform database schema is not provided.

APPROACHING COMPETENCE

A screenshot of a Google Cloud Platform database schema is provided but does not contain *all* appropriate schema objects, or *all* attributes from the dataset are not provided.

COMPETENT

A screenshot showing a Google Cloud Platform database schema is provided and contains *all* appropriate schema objects that incorporate *all* attributes from the dataset.

A1A:GCP DATABASE SCHEMA DOCUMENTATION

NOT EVIDENT

Documentation of code used to create the database schema is not submitted

APPROACHING COMPETENCE

Documentation is submitted explaining any code used to create the data schema in part A1, but

COMPETENT

Documentation is submitted identifying and explaining the function of all the components of the code used to create the data schema in part A1.

does not identify code components, or how they function.

B1:POPULATION OF DATABASE

NOT EVIDENT

A screenshot showing the inserted records is not provided.

APPROACHING COMPETENCE

A screenshot from the GCP database shows fewer than 10 inserted transactions, or the transaction records are not from the sample transaction dataset.

COMPETENT

A screenshot from the GCP database showing *at least* 10 inserted transaction records from the sample transaction dataset is provided.

B2:CODE COMPONENTS

NOT EVIDENT

Documentation and an explanation of the components of the code used in part B1 are not provided.

APPROACHING COMPETENCE

Documentation is provided but an explanation of code components or functions is missing. Or an explanation of the components of code used in part B1 is provided but is missing either documentation or an explanation of how they function.

COMPETENT

Documentation and an explanation of the components of the code used to insert records in part B1 and how they function is provided.

C:PANOPTO RECORDING

NOT EVIDENT

The submission does not provide a Panopto video recording.

APPROACHING COMPETENCE

The submission provides a Panopto video recording that includes either a screen share of the presenter demonstrating the queries within the database, or a verbal explanation of how the queries were written.

COMPETENT

The submission provides a Panopto video recording that includes *both* a screen share of the presenter demonstrating the queries running within the database and a verbal explanation of how they are written.

C1:UNIQUE CUSTOMER QUERY

NOT EVIDENT

The submission does not include a query listing the unique cus-

APPROACHING COMPETENCE

COMPETENT

The submission includes a query listing *all* the unique customers in

tomers in the database.

The submission includes a query listing some unique customers in the database but is incomplete.

the database.

C2: CUSTOMER SHOPPING CART

NOT EVIDENT

The submission does not include a query listing the items in a single customer's shopping cart.

APPROACHING COMPETENCE

The submission includes a query listing some items in a single customer's shopping cart but is incomplete.

COMPETENT

The submission includes a query listing *all* the items in a single customer's shopping cart.

C3: TOTAL PURCHASE AMOUNTS

NOT EVIDENT

The submission does not include a query listing the total purchase amounts for customers in descending order.

APPROACHING COMPETENCE

The submission includes a query listing the total purchase amounts for some customers but not all, or the amounts are not in descending order.

COMPETENT

The submission includes a query listing the total purchase amounts for *all* customers in the database in descending order.

D: SOURCES

NOT EVIDENT

The submission does not include both in-text citations and a reference list for sources that are quoted, paraphrased, or summarized.

APPROACHING COMPETENCE

The submission includes in-text citations for sources that are quoted, paraphrased, or summarized and a reference list; however, the citations or reference list is incomplete or inaccurate.

COMPETENT

The submission includes in-text citations for sources that are properly quoted, paraphrased, or summarized and a reference list that accurately identifies the author, date, title, and source location as available or the submission states no sources were used.

E: PROFESSIONAL COMMUNICATION

NOT EVIDENT

This submission includes pervasive errors in professional communication related to grammar, sentence fluency, contextual spelling, or punctuation, nega-

APPROACHING COMPETENCE

This submission includes substantial errors in professional communication related to grammar, sentence fluency, contextu-

COMPETENT

This submission includes satisfactory use of grammar, sentence fluency, contextual spelling, and punctuation, which promote ac-

tively impacting the professional quality and clarity of the writing. Specific errors have been identified by Grammarly for Education under the Correctness category.

al spelling, or punctuation. Specific errors have been identified by Grammarly for Education under the Correctness category.

curate interpretation and understanding.

WEB LINKS

[Panopto Access](#)

Sign in using the "WGU" option. If prompted, log in with your WGU student portal credentials, which should forward you to Panopto's website. If you have any problems accessing Panopto, please contact Assessment Services at assessmentservices@wgu.edu. It may take up to two business days to receive your WGU Panopto recording permissions once you have begun the course.

[Panopto FAQs](#)

[Panopto How-To Videos](#)

[WGU GCP database Playground](#)

[Databases DPN2 | D607](#)

SUPPORTING DOCUMENTS

[Alliah Company Scenario.docx](#)

[Data Dictionary for Transaction Data.md](#)

[TransactionData.json](#)